

# Literature Answer: Compact Manifold CLSI

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## Status

**Status:** `literature_already_answered`.

Problem 1 from the final “Major Open Problems” list in arXiv:1807.08838 is answered by arXiv:2007.06138.

## Source Question

The source is Li Gao, Marius Junge, and Nicholas LaRacuente, *Fisher Information and Logarithmic Sobolev Inequality for Matrix Valued Functions*, arXiv:1807.08838. In the subsection “Major Open Problems”, the authors ask:

Does every compact Riemannian manifold satisfy CLSI?

This is the first of three questions in that final problem list.

## Supporting Answer

The supporting paper is Michael Brannan, Li Gao, and Marius Junge, *Complete logarithmic Sobolev inequalities via Ricci curvature bounded below*, arXiv:2007.06138, published in *Advances in Mathematics* 394 (2022), article 108129.

The supporting paper cites arXiv:1807.08838 under the bibliography key CLSI. Its introduction states that every heat semigroup on a connected compact weighted Riemannian manifold satisfies CLSI. In the arXiv source, Theorem `riemann` proves the stronger statement: if  $(M, g, e^{-W} d\text{vol})$  is a compact connected weighted Riemannian manifold, then the weighted heat semigroup

$$T_t = e^{-t\Delta_W}$$

satisfies  $\lambda$ -CLSI for some  $\lambda > 0$ . Taking  $W = 0$  gives the Laplace-Beltrami heat semigroup on every compact connected Riemannian manifold, which is exactly the source question.

## Scope Limitations

Only Problem 1 is marked answered here. The source also asks whether every generator of a selfadjoint semigroup on a matrix algebra satisfies CLSI, and whether CLSI is stable under free products. Those two questions are not resolved by this packet.

## Evidence Used

The local source file for arXiv:1807.08838 contains the final “Major Open Problems” section. The arXiv source bundle for arXiv:2007.06138 contains the citation to arXiv:1807.08838 as CLSI, the introductory compact-manifold claim, and Theorem **riemann**. The same result is visible in public article metadata and abstract records for the Advances in Mathematics paper.